

$$\lim_{\alpha\rightarrow\beta\rightarrow0}G_X=\frac{1}{2}$$

Following are the limits with one parameter finite (non-zero) and the other approaching these limits:

$$\lim_{\beta\rightarrow\infty}G_X=\lim_{\alpha\rightarrow\infty}G_X=1$$

$$\lim_{\alpha\rightarrow0}G_X=\lim_{\beta\rightarrow0}G_X=0$$

The accompanying plot shows the difference between the mean and the geometric mean for shape parameters α and β from zero to 2. Besides the fact that the difference becomes large for values of α and β approaching zero, one can observe an evident asymmetry of the geometric mean with respect to the shape parameters α and β. The difference between the geometric mean and the mean is larger for small values of α in relation to β than when exchanging the magnitudes of β and α.

N. L.Johnson and S. Kotz^[3] suggest the logarithmic approximation to the digamma function ψ(n) = ln(α − 1/2) which results in the following approximation to the geometric mean:

$$G_X\approx\frac{\alpha-\frac{1}{2}}{\alpha+\beta-\frac{1}{2}}\text{ if }\alpha,\beta>1.$$

Numerical values for the relative error in this approximation follow: [(α = β = 1): 9.39%]; [(α = β = 2): 1.29%]; [(α = 2, β = 2): 1.51%]; [(α = 3, β = 2): 0.44%]; [(α = β = 3): 0.51%]; [(α = 3, β = 4): 0.26%]; [(α = 4, β = 2): 0.24%].

Similarly, one can calculate the value of shape parameters required for the geometric mean to equal 1/2. Given the value of the parameter β, what would be the value of the other parameter, α, required for the geometric mean to equal 1/2? The answer is that (for β > 1), the value of α required tends towards β + 1/2 as β → ∞. For example, all these couples have the same geometric mean of 1/2: [β = 1, α = 1.4427], [β = 2, α = 2.46938], [β = 3, α = 3.47943], [β = 4, α = 4.48449], [β = 5, α = 5.48756], [β = 10, α = 10.4938], [β = 100, α = 100.499].

The fundamental property of the geometric mean, which can be proven to be false for any other mean, is

$$G\left(\frac{X_i}{Y_i}\right)=\frac{G(X_i)}{G(Y_i)}$$

This makes the geometric mean the only correct mean when averaging normalized results, that is results that are presented as ratios to reference values.^[3] This is relevant because the beta distribution is a suitable model for the random behavior of percentages and it is particularly suitable to the statistical modelling of proportions. The geometric mean plays a central role in maximum likelihood estimation, see section "Parameter estimation, maximum likelihood." Actually, when performing maximum likelihood estimation, besides the geometric mean G*G* based on the random variable X, also another geometric mean appears naturally: the *geometric mean* based on the linear transformation −(1 − X), the mirror-image of X, denoted by G*G*_{1−*X*}.

$$G_{(1-X)}=e^{\psi(\ln(1-X))}=e^{\psi(\beta)-\psi(\alpha+\beta)}$$

Along a line α = β, the following limits apply:

$$\lim_{\alpha\rightarrow\beta\rightarrow0}G_{(1-X)}=0$$

$$\lim_{\alpha\rightarrow\beta\rightarrow\infty}G_{(1-X)}=\frac{1}{2}$$

Following are the limits with one parameter finite (non-zero) and the other approaching these limits:

$$\lim_{\beta\rightarrow\infty}G_{(1-X)}=\lim_{\alpha\rightarrow\infty}G_{(1-X)}=0$$

$$\lim_{\alpha\rightarrow0}G_{(1-X)}=\lim_{\beta\rightarrow0}G_{(1-X)}=1$$

It has the following approximate value:

$$G_{(1-X)}\approx\frac{\beta-\frac{1}{2}}{\alpha+\beta-\frac{1}{2}}\text{ if }\alpha,\beta>1.$$

Although both G*G* and G*G*_{1−*X*} are asymmetric, in the case that both shape parameters are equal α = β, the geometric means are equal: G*G* = G*G*_{1−*X*}. This equality follows from the following symmetry displayed between both geometric means:

$$G_X(\mathbf{B}(\alpha,\beta))=G_{1-X}(\mathbf{B}(\beta,\alpha)).$$

Harmonic mean

The inverse of the harmonic mean (H*H*) of a distribution with random variable X is the arithmetic mean of 1/X, or, equivalently, its expected value. Therefore, the *harmonic mean* (H*H*) of a beta distribution with shape parameters α and β is:

$$\begin{aligned} H_X&=\frac{1}{\mathbb{E}\left[\frac{1}{X}\right]}\\ &=\frac{1}{\int_0^1\frac{t^{\alpha-1}(1-t)^{\beta-1}}{1-t}dt}\\ &=\frac{1}{\int_0^1\frac{t^{\alpha-1}(1-t)^{\beta-1}}{-\ln(t)}dt}\\ &=\frac{\alpha-1}{\alpha+\beta-1}\text{ if }\alpha>1\text{ and }\beta>0 \end{aligned}$$

The harmonic mean (H*H*) of a beta distribution with α < 1 is undefined, because its defining expression is not bounded in [0, 1] for shape parameter α less than unity.

Letting α = β in the above expression one obtains

$$H_X=\frac{\alpha-1}{2\alpha-1},$$

showing that for α = β the harmonic mean ranges from 0, for α = β = 1, to 1/2, for α = β = ∞.

Following are the limits with one parameter finite (non-zero) and the other approaching these limits:

$$\begin{aligned} \lim_{\alpha\rightarrow0}H_X&\text{ is undefined} \\ \lim_{\alpha\rightarrow1}H_X&=\lim_{\beta\rightarrow0}H_X=0 \\ \lim_{\beta\rightarrow\infty}H_X&=\lim_{\alpha\rightarrow\infty}H_X=1 \end{aligned}$$

The harmonic mean plays a role in maximum likelihood estimation for the four parameter case, in addition to the geometric mean. Actually, when performing maximum likelihood estimation for the four parameter case, besides the harmonic mean H*H* based on the random variable X, also another harmonic mean appears naturally: the harmonic mean based on the linear transformation (1 − X), the mirror-image of X, denoted by H*H*_{1−*X*}:

$$H_{1-X}=\frac{1}{\mathbb{E}\left[\frac{1}{1-X}\right]}=\frac{\beta-1}{\alpha+\beta-1}\text{ if }\beta>1,\text{ and }\alpha>0.$$

The harmonic mean (H*H*_{1−*X*}) of a beta distribution with β < 1 is undefined, because its defining expression is not bounded in [0, 1] for shape parameter β less than unity.

Letting α = β in the above expression one obtains

$$H_{(1-X)}=\frac{\beta-1}{2\beta-1},$$

showing that for α = β the harmonic mean ranges from 0, for α = β = 1, to 1/2, for α = β = ∞.

Following are the limits with one parameter finite (non-zero) and the other approaching these limits:

$$\begin{aligned} \lim_{\beta\rightarrow0}H_{1-X}&\text{ is undefined} \\ \lim_{\beta\rightarrow1}H_{1-X}&=\lim_{\alpha\rightarrow0}H_{1-X}=0 \\ \lim_{\alpha\rightarrow\infty}H_{1-X}&=\lim_{\beta\rightarrow\infty}H_{1-X}=1 \end{aligned}$$

Although both H*H* and H*H*_{1−*X*} are asymmetric, in the case that both shape parameters are equal α = β, the harmonic means are equal: H*H* = H*H*_{1−*X*}. This equality follows from the following symmetry displayed between both harmonic means:

$$H_X(\mathbf{B}(\alpha,\beta))=H_{1-X}(\mathbf{B}(\beta,\alpha))\text{ if }\alpha,\beta>1.$$

Measures of statistical dispersion

Variance

The *variance* (the second moment centered on the mean) of a beta distribution random variable X with parameters α and β is:^{[3][3]}

$$\mathbf{var}(X)=\mathbb{E}[(X-\mu)^2]=\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$$

Letting α = β in the above expression one obtains

$$\mathbf{var}(X)=\frac{1}{4(2\beta+1)},$$

showing that for α = β the variance decreases monotonically as α = β increases. Setting α = β = 0 in this expression, one finds the maximum variance var(X) = 1/4^[3] which only occurs approaching the limit, at α = β = 0.

The beta distribution may also be parametrized in terms of its mean μ (0 < μ < 1) and sample size v = α + β (v > 0) (see subsection Mean and sample size):

$$\begin{aligned} \alpha&=\mu v,\text{ where }\nu=(\alpha+\beta)>0 \\ \beta&=(1-\mu)v,\text{ where }\nu=(\alpha+\beta)>0. \end{aligned}$$

Using this parametrization, one can express the variance in terms of the mean μ and the sample size v as follows:

$$\mathbf{var}(X)=\frac{\mu(1-\mu)}{1+\nu}$$

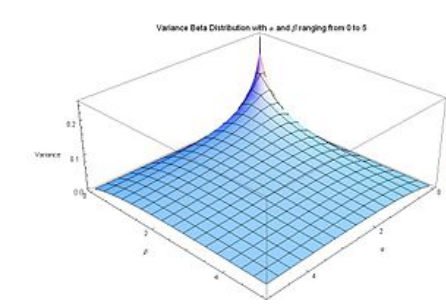
Since v = α + β > 0, it follows that var(X) < μ(1 − μ).

For a symmetric distribution, the mean is at the middle of the distribution, μ = 1/2, and therefore:

$$\mathbf{var}(X)=\frac{1}{4(1+\nu)}\text{ if }\mu=\frac{1}{2}$$

Also, the following limits (with only the noted variable approaching the limit) can be obtained from the above expressions:

$$\begin{aligned} \lim_{\beta\rightarrow0}\mathbf{var}(X)&=\lim_{\alpha\rightarrow0}\mathbf{var}(X)=\lim_{\beta\rightarrow\infty}\mathbf{var}(X)=\lim_{\alpha\rightarrow\infty}\mathbf{var}(X)=\lim_{\mu\rightarrow0}\mathbf{var}(X)=\lim_{\mu\rightarrow1}\mathbf{var}(X)=0 \\ \lim_{\beta\rightarrow0}\mathbf{var}(X)&=\mu(1-\mu) \end{aligned}$$



Geometric variance and covariance

The logarithm of the geometric variance, ln(var*G*_G), of a distribution with random variable X is the second moment of the logarithm of X centered on the geometric mean of X, ln(G*G*_G):

$$\begin{aligned} \ln\mathbf{var}_{GX}&=\mathbb{E}[(\ln X-\ln G_X)^2] \\ &=\mathbb{E}[(\ln X-\mathbb{E}[\ln X])^2] \\ &=\mathbb{E}[(\ln X)^2]-\left(\mathbb{E}[\ln X]\right)^2 \\ &=\mathbf{var}[\ln X] \end{aligned}$$

and therefore, the geometric variance is:

$$\mathbf{var}_{GX}=e^{\mathbf{var}[\ln X]}$$

In the Fisher information matrix, and the curvature of the log likelihood function, the logarithm of the geometric variance of the *reflected* variable 1 − X and the logarithm of the geometric covariance between X and 1 − X appear:

$$\begin{aligned} \ln\mathbf{var}_{G(1-X)}&=\mathbb{E}[(\ln(1-X)-\ln G_{1-X})^2] \\ &=\mathbb{E}[(\ln(1-X)-\mathbb{E}[\ln(1-X)])^2] \\ &=\mathbb{E}[(\ln(1-X))^2]-\left(\mathbb{E}[\ln(1-X)]\right)^2 \\ &=\mathbf{var}[\ln(1-X)] \end{aligned}$$

$$\mathbf{var}_{G(1-X)}=e^{\mathbf{var}[\ln(1-X)]}$$

$$\ln\mathbf{cov}_{GX,1-X}=\mathbb{E}[(\ln X-\ln G_X)(\ln(1-X)-\ln G_{1-X})]$$

$$\begin{aligned} &=\mathbb{E}[(\ln X-\mathbb{E}[\ln X])(\ln(1-X)-\mathbb{E}[\ln(1-X)])] \\ &=\mathbb{E}[\ln X\ln(2-X)]-\mathbb{E}[\ln X]\mathbb{E}[\ln(1-X)] \\ &=\mathbf{cov}[\ln X,\ln(1-X)] \end{aligned}$$

$$\mathbf{cov}_{GX,(1-X)}=e^{\mathbf{cov}[\ln X,\ln(1-X)]}$$

For a beta distribution, higher order logarithmic moments can be derived by using the representation of a beta distribution as a proportion of two gamma distributions and differentiating through the integral. They can be expressed in terms of higher order poly-gamma functions. See the section § Moments of logarithmically transformed random variables. The variance of the logarithmic variables and covariance of ln X and ln(1 − X) are:

$$\begin{aligned} \mathbf{var}[\ln X]&=\psi_1(\alpha)-\psi_1(\alpha+\beta) \\ \mathbf{var}[\ln(1-X)]&=\psi_1(\beta)-\psi_1(\alpha+\beta) \\ \mathbf{cov}[\ln X,\ln(1-X)]&=-\psi_1(\alpha+\beta) \end{aligned}$$

where the **trigamma function**, denoted ψ*ψ*(α), is the second of the polygamma functions, and is defined as the derivative of the digamma function:

$$\psi_1(\alpha)=\frac{d^2}{d\alpha^2}\ln\Gamma(\alpha)=\frac{d}{d\alpha}\psi(\alpha).$$

Therefore,

$$\begin{aligned} \ln\mathbf{var}_{GX}&=\mathbf{var}[\ln X]=\psi_1(\alpha)-\psi_1(\alpha+\beta) \\ \ln\mathbf{var}_{G(1-X)}&=\mathbf{var}[\ln(1-X)]=\psi_1(\beta)-\psi_1(\alpha+\beta) \\ \ln\mathbf{cov}_{GX,1-X}&=\mathbf{cov}[\ln X,\ln(1-X)]=-\psi_1(\alpha+\beta) \end{aligned}$$

The accompanying plots show the log geometric variances and log geometric covariance versus the shape parameters α and β. The plots show that the log geometric variances and log geometric covariance are close to zero for shape parameters α and β greater than 2, and that the log geometric variances rapidly rise in value for shape parameter values α and β less than unity. The log geometric variances are positive for all values of the shape parameters. The log geometric covariance is negative for all values of the shape parameters, and it reaches large negative values for α and β less than unity.

Following are the limits with one parameter finite (non-zero) and the other approaching these limits:

$$\begin{aligned} \lim_{\alpha\rightarrow0}\ln\mathbf{var}_{GX}&=\lim_{\beta\rightarrow0}\ln\mathbf{var}_{G(1-X)}=-\infty \\ \lim_{\beta\rightarrow\infty}\ln\mathbf{var}_{GX}&=\lim_{\alpha\rightarrow\infty}\ln\mathbf{var}_{GX}=\lim_{\beta\rightarrow0}\ln\mathbf{var}_{G(1-X)}=\lim_{\alpha\rightarrow0}\ln\mathbf{var}_{G(1-X)}=\lim_{\mu\rightarrow0}\ln\mathbf{cov}_{GX,(1-X)}=\lim_{\mu\rightarrow1}\ln\mathbf{cov}_{GX,(1-X)}=0 \\ \lim_{\alpha\rightarrow0}\ln\mathbf{var}_{GX}&=\psi_1(\alpha) \\ \lim_{\alpha\rightarrow\infty}\ln\mathbf{var}_{G(1-X)}&=\psi_1(\beta) \\ \lim_{\alpha\rightarrow\infty}\ln\mathbf{cov}_{GX,(1-X)}&=-\psi_1(\beta) \\ \lim_{\beta\rightarrow0}\ln\mathbf{cov}_{GX,(1-X)}&=-\psi_1(\alpha) \end{aligned}$$

Limits with two parameters varying:

$$\begin{aligned} \lim_{\alpha\rightarrow\infty}\left(\lim_{\beta\rightarrow\infty}\ln\mathbf{var}_{GX}\right)&=\lim_{\beta\rightarrow\infty}\left(\lim_{\alpha\rightarrow\infty}\ln\mathbf{var}_{G(1-X)}\right)=\lim_{\alpha\rightarrow\infty}\left(\lim_{\beta\rightarrow\infty}\ln\mathbf{cov}_{GX,(1-X)}\right)=\lim_{\beta\rightarrow\infty}\left(\lim_{\alpha\rightarrow\infty}\ln\mathbf{cov}_{GX,(1-X)}\right)=0 \\ \lim_{\alpha\rightarrow\infty}\left(\lim_{\beta\rightarrow0}\ln\mathbf{var}_{GX}\right)&=\lim_{\beta\rightarrow0}\left(\lim_{\alpha\rightarrow\infty}\ln\mathbf{var}_{G(1-X)}\right)=-\infty \\ \lim_{\alpha\rightarrow\infty}\left(\lim_{\beta\rightarrow0}\ln\mathbf{cov}_{GX,(1-X)}\right)&=\lim_{\beta\rightarrow0}\left(\lim_{\alpha\rightarrow\infty}\ln\mathbf{cov}_{GX,(1-X)}\right)=-\infty \end{aligned}$$

Although both ln(var*G*_G) and ln(var*G*_{1−*X*}) are asymmetric, when the shape parameters are equal, α = β, one has: ln(var*G*_G) = ln(var*G*_{1−*X*}). This equality follows from the following symmetry displayed between both log geometric variances:

$$\ln\mathbf{var}_{GX}(\mathbf{B}(\alpha,\beta))=\ln\mathbf{var}_{G(1-X)}(\mathbf{B}(\beta,\alpha)).$$

The log geometric covariance is symmetric:

$$\ln\mathbf{cov}_{GX,(1-X)}(\mathbf{B}(\alpha,\beta))=\ln\mathbf{cov}_{GX,(1-X)}(\mathbf{B}(\beta,\alpha))$$

Mean absolute deviation around the mean

The mean absolute deviation around the mean for the beta distribution with shape parameters α and β is:^[8]

$$\mathbb{E}[|X-E[X]|]=\frac{2\alpha^\alpha\beta^\beta}{\mathbf{B}(\alpha,\beta)(\alpha+\beta)^{\alpha+\beta+1}}$$

The mean absolute deviation around the mean is a more robust estimator of statistical dispersion than the standard deviation for beta distributions with tails and inflection points at each side of the mode, Beta(α, β) distributions with α, β > 2, as it depends on the linear (absolute) deviations rather than the square deviations from the mean. Therefore, the effect of very large deviations from the mean are not as overly weighted.

Using Stirling's approximation to the Gamma function, N.L.Johnson and S.Kotz^[3] derived the following approximation for values of the shape parameters greater than unity (the relative error for this approximation is only ~3.5% for α = β = 1, and it decreases to zero as α → ∞, β → ∞):

$$\begin{aligned} \frac{\text{mean abs. dev. from mean}}{\text{standard deviation}}&=\frac{\mathbb{E}[|X-E[X]|]}{\sqrt{\mathbf{var}(X)}}\\ &\approx\sqrt{\frac{2}{\pi}}\left(1+\frac{7}{12(\alpha+\beta)}-\frac{1}{12\alpha}-\frac{1}{12\beta}\right),\text{ if }\alpha,\beta>1. \end{aligned}$$

At the limit α → ∞, β → ∞, the ratio of the mean absolute deviation to the standard deviation (for the beta distribution) becomes equal to the ratio of the same measures for the normal distribution: 2√2⁄π. For α = β = 1 this ratio equals 2√8⁄π, so that from α = β = 1 to α = β → ∞ the ratio decreases by 8.5%. For α = β = 0 the standard deviation is exactly equal to the mean absolute deviation around the mean. Therefore, this ratio decreases by 15% from α = β = 0 to α = β = 1, and by 25% from α = β = 0 to α = β → ∞. However, for skewed beta distributions such that α → 0 or β → 0, the ratio of the standard deviation to the mean absolute deviation approaches infinity (although each of them, individually, approaches zero) faster than the standard deviation.

Using the parametrization in terms of mean μ and sample size v = α + β > 0:

$$\alpha=\mu v,\beta=(1-\mu)v$$

one can express the mean absolute deviation around the mean in terms of the mean μ and the sample size v as follows:

$$\mathbb{E}[|X-E[X]|]=\frac{2\mu^\mu(1-\mu)^{1-\mu}}{v\mathbf{B}(\mu\nu,(1-\mu)\nu)}$$

For a symmetric distribution, the mean is at the middle of the distribution, μ = 1/2, and therefore:

$$\mathbb{E}[|X-E[X]|]=\frac{2^{1-\nu}}{v\mathbf{B}(\frac{1}{2},\frac{1}{2})}=\frac{2^{1-\nu}\Gamma(\nu)}{v\Gamma(\frac{1}{2})^2}$$

$$\lim_{\nu\rightarrow1}\left(\lim_{\mu\rightarrow\frac{1}{2}}\mathbb{E}[|X-E[X]|]\right)=\frac{1}{2}$$

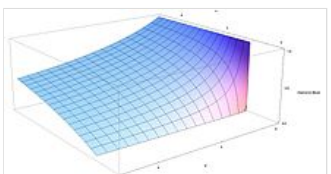
$$\lim_{\nu\rightarrow1}\left(\lim_{\mu\rightarrow1}\mathbb{E}[|X-E[X]|]\right)=0$$

Also, the following limits (with only the noted variable approaching the limit) can be obtained from the above expressions:

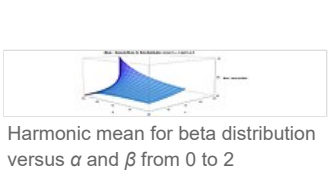
$$\begin{aligned} \lim_{\mu\rightarrow0}\mathbb{E}[|X-E[X]|]&=\lim_{\mu\rightarrow0}\mathbb{E}[|X-E[X]|]=0 \\ \lim_{\mu\rightarrow1}\mathbb{E}[|X-E[X]|]&=\lim_{\mu\rightarrow1}\mathbb{E}[|X-E[X]|]=0 \\ \lim_{\mu\rightarrow0}\mathbb{E}[|X-E[X]|]&=\lim_{\mu\rightarrow0}\mathbb{E}[|X-E[X]|]=0 \\ \lim_{\mu\rightarrow1}\mathbb{E}[|X-E[X]|]&=\sqrt{\mu(1-\mu)} \\ \lim_{\nu\rightarrow\infty}\mathbb{E}[|X-E[X]|]&=0 \end{aligned}$$

Mean absolute difference

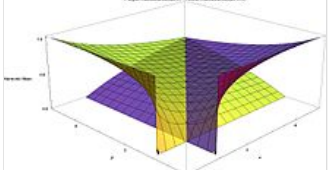
The mean absolute difference for the beta distribution is:



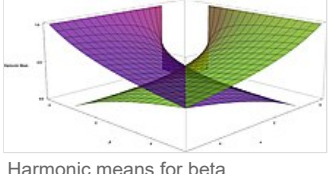
Harmonic mean for beta distribution for 0 < α < 5 and 0 < β < 5



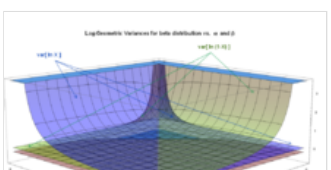
Harmonic mean for beta distribution versus α and β from 0 to 2



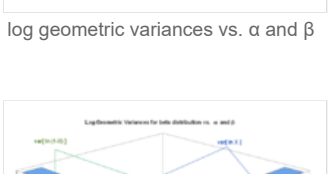
Harmonic means for beta distribution: Purple = H(X), Yellow = H(1−X), smaller values α and β in front



Harmonic means for beta distribution: purple = H(X), yellow = H(1−X), larger values α and β in front



log geometric variances vs. α and β



log geometric variances vs. α and β

Moments of logarithmically transformed random variables

Expected values for logarithmic transformations (useful for maximum likelihood estimates, see § Parameter estimation, Maximum likelihood) are discussed in this section. The following logarithmic linear transformations are related to the geometric means G_X and G_{1-X} (see § Geometric Mean):

E[ln(X)] = ψ(α) - ψ(α + β) = -E[ln(1/X)]
E[ln(1 - X)] = ψ(β) - ψ(α + β) = -E[ln(1/(1-X))]

Where the digamma function ψ(α) is defined as the logarithmic derivative of the gamma function.[27]

ψ(α) = d ln Γ(α) / dα

Logit transformations are interesting,[28] as they usually transform various shapes (including J-shapes) into (usually skewed) bell-shaped densities over the logit variable, and they may remove the end singularities over the original variable:

E[ln(X/(1-X))] = ψ(α) - ψ(β) = E[ln(X)] + E[ln(1/(1-X))]
E[ln(1-X/X)] = ψ(β) - ψ(α) = -E[ln(X/(1-X))]

Johnson[28] considered the distribution of the logit - transformed variable ln(X/(1 - X)), including its moment generating function and approximations for large values of the shape parameters. This transformation extends the finite support [0, 1] based on the original variable X to infinite support in both directions of the real line (-∞, +∞). The logit of a beta variate has the logistic-beta distribution.

Higher order logarithmic moments can be derived by using the representation of a beta distribution as a proportion of two gamma distributions and differentiating through the integral. They can be expressed in terms of higher order poly-gamma functions as follows:

E[ln^2(X)] = (ψ(α) - ψ(α + β))^2 + ψ_1(α) - ψ_1(α + β)
E[ln^2(1 - X)] = (ψ(β) - ψ(α + β))^2 + ψ_1(β) - ψ_1(α + β)
E[ln(X) ln(1 - X)] = (ψ(α) - ψ(α + β))(ψ(β) - ψ(α + β)) - ψ_1(α + β)

therefore the variance of the logarithmic variables and covariance of ln(X) and ln(1-X) are:

cov[ln(X), ln(1 - X)] = E[ln(X) ln(1 - X)] - E[ln(X)] E[ln(1 - X)] = -ψ_1(α + β)
var[ln X] = E[ln^2(X)] - (E[ln(X)])^2 = ψ_1(α) - ψ_1(α + β) = ψ_1(α) + cov[ln(X), ln(1 - X)]
var[ln(1 - X)] = E[ln^2(1 - X)] - (E[ln(1 - X)])^2 = ψ_1(β) - ψ_1(α + β) = ψ_1(β) + cov[ln(X), ln(1 - X)]

where the trigamma function, denoted ψ_1(α), is the second of the polygamma functions, and is defined as the derivative of the digamma function:

ψ_1(α) = d^2 ln Γ(α) / dα^2 = -dψ(α) / dα

The variances and covariance of the logarithmically transformed variables X and (1 - X) are different, in general, because the logarithmic transformation destroys the mirror-symmetry of the original variables X and (1 - X), as the logarithm approaches negative infinity for the variable approaching zero.

These logarithmic variances and covariance are the elements of the Fisher information matrix for the beta distribution. They are also a measure of the curvature of the log likelihood function (see section on Maximum likelihood estimation).

The variances of the log inverse variables are identical to the variances of the log variables:

var[ln(1/X)] = var[ln(X)] = ψ_1(α) - ψ_1(α + β)
var[ln(1/(1-X))] = var[ln(1 - X)] = ψ_1(β) - ψ_1(α + β)
cov[ln(1/X), ln(1/(1-X))] = cov[ln(X), ln(1 - X)] = -ψ_1(α + β)

It also follows that the variances of the logit-transformed variables are

var[ln(X/(1-X))] = var[ln(1-X/X)] = -cov[ln(X/(1-X)), ln(1-X/X)] = ψ_1(α) + ψ_1(β).

Quantities of information (entropy)

Given a beta distributed random variable, X ~ Beta(α, β), the differential entropy of X (is measured in nats),[29] the expected value of the negative of the logarithm of the probability density function:

h(X) = E[-ln(f(x; α, β))]
= ∫_0^1 -f(x; α, β) ln(f(x; α, β)) dx
= ln(B(α, β)) - (α - 1)ψ(α) - (β - 1)ψ(β) + (α + β - 2)ψ(α + β)

where f(x; α, β) is the probability density function of the beta distribution:

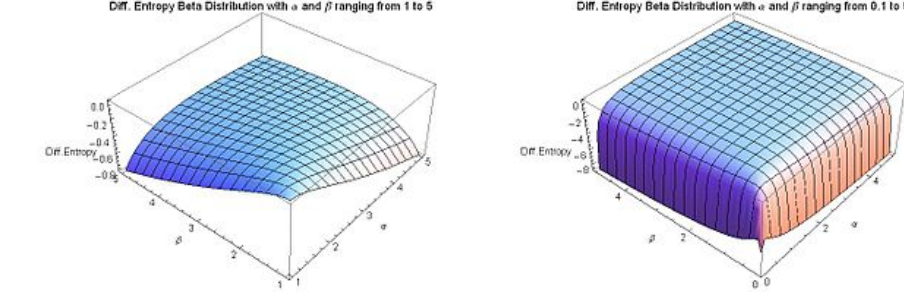
f(x; α, β) = 1/B(α, β) * x^{α-1} * (1-x)^{β-1}

The digamma function ψ appears in the formula for the differential entropy as a consequence of Euler's integral formula for the harmonic numbers which follows from the integral:

∫_0^1 1-x^{p-1} / (1-x) dx = ψ(p) - ψ(1)

The differential entropy of the beta distribution is negative for all values of α and β greater than zero, except at α = β = 1 (for which values the beta distribution is the same as the uniform distribution), where the differential entropy reaches its maximum value of zero. It is to be expected that the maximum entropy should take place when the beta distribution becomes equal to the uniform distribution, since uncertainty is maximal when all possible events are equiprobable.

For α or β approaching zero, the differential entropy approaches its minimum value of negative infinity. For (either or both) α or β approaching zero, there is a maximum amount of order: all the probability density is concentrated at the ends, and there is zero probability density at points located between the ends. Similarly for (either or both) α or β approaching infinity, the differential entropy approaches its minimum value of negative infinity, and a maximum amount of order. If either α or β approaches infinity (and the other is finite) all the probability density is concentrated at an end, and the probability density is zero everywhere else. If both shape parameters are equal (the symmetric case), α = β, and they approach infinity simultaneously, the probability density becomes a spike (Dirac delta function) concentrated at the middle x = 1/2, and hence there is 100% probability at the middle x = 1/2 and zero probability everywhere else.



The (continuous case) differential entropy was introduced by Shannon in his original paper (where he named it the "entropy of a continuous distribution"), as the concluding part of the same paper where he defined the discrete entropy.[26] It is known since then that the differential entropy may differ from the infinitesimal limit of the discrete entropy by an infinite offset, therefore the differential entropy can be negative (as it is for the beta distribution). What really matters is the relative value of entropy.

Given two beta distributed random variables, X_1 ~ Beta(α, β) and X_2 ~ Beta(α', β'), the cross-entropy is (measured in nats)[27]

H(X_1, X_2) = ∫_0^1 -f(x; α, β) ln(f(x; α', β')) dx
= ln(B(α', β')) - (α' - 1)ψ(α') - (β' - 1)ψ(β') + (α' + β' - 2)ψ(α' + β).

The cross entropy has been used as an error metric to measure the distance between two hypotheses.[28][29] Its absolute value is minimum when the two distributions are identical. It is the information measure most closely related to the log maximum likelihood [27](see section on "Parameter estimation, Maximum likelihood estimation").

The relative entropy, or Kullback–Leibler divergence D_KL(X_1 || X_2), is a measure of the inefficiency of assuming that the distribution is X_2 ~ Beta(α', β') when the distribution is really X_1 ~ Beta(α, β). It is defined as follows (measured in nats).

D_KL(X_1 || X_2) = ∫_0^1 f(x; α, β) ln(f(x; α, β) / f(x; α', β')) dx
= (∫_0^1 f(x; α, β) ln(f(x; α, β)) dx) - (∫_0^1 f(x; α, β) ln(f(x; α', β')) dx)
= -h(X_1) + H(X_1, X_2)
= ln(B(α', β') / B(α, β)) + (α - α')ψ(α) + (β - β')ψ(β) + (α' - α + β' - β)ψ(α + β).

The relative entropy, or Kullback–Leibler divergence, is always non-negative. A few numerical examples follow:

- X_1 ~ Beta(1, 1) and X_2 ~ Beta(3, 3): D_KL(X_1 || X_2) = 0.598903; D_KL(X_2 || X_1) = 0.267864; h(X_1) = 0; h(X_2) = -0.267864
- X_1 ~ Beta(α, 0.5) and X_2 ~ Beta(0.5, 3): D_KL(X_1 || X_2) = 7.21574; D_KL(X_2 || X_1) = 7.21574; h(X_1) = -1.10805; h(X_2) = -1.10805.

The Kullback–Leibler divergence is not symmetric D_KL(X_1 || X_2) ≠ D_KL(X_2 || X_1) for the case in which the individual beta distributions Beta(1, 1) and Beta(3, 3) are symmetric, but have different entropies h(X_1) > h(X_2). The value of the Kullback divergence depends on the direction traveled: whether going from a higher (differential) entropy to a lower (differential) entropy or the other way around. In the numerical example above, the Kullback divergence measures the inefficiency of assuming that the distribution is (bell-shaped) Beta(3, 3), rather than (uniform) Beta(1, 1). The "h" entropy of Beta(1, 1) is higher than the "h" entropy of Beta(3, 3) because the uniform distribution Beta(1, 1) has a maximum amount of disorder. The Kullback divergence is more than two times higher (0.598903 instead of 0.267864) when measured in the direction of decreasing entropy: the direction that assumes that the (uniform) Beta(1, 1) distribution is (bell-shaped) Beta(3, 3) rather than the other way around. In this restricted sense, the Kullback divergence is consistent with the second law of thermodynamics.

The Kullback–Leibler divergence is symmetric D_KL(X_1 || X_2) = D_KL(X_2 || X_1) for the skewed cases Beta(3, 0.5) and Beta(0.5, 3) that have equal differential entropy h(X_1) = h(X_2).

The symmetry condition:

D_KL(X_1 || X_2) = D_KL(X_2 || X_1), if h(X_1) = h(X_2), for (skewed) α ≠ β

follows from the above definitions and the mirror-symmetry f(x; α, β) = f(1 - x; α, β) enjoyed by the beta distribution.

Relationships between statistical measures

Mean, mode and median relationship

If 1 < α < β then mode < median < mean.[3] Expressing the mode (only for α, β > 1), and the mean in terms of α and β:

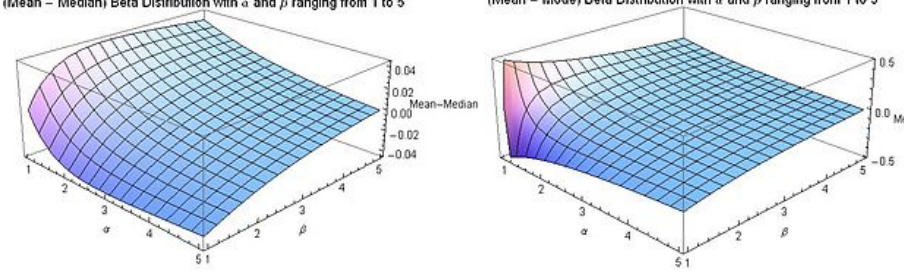
(α-1)/(α+β-2) ≤ median ≤ α/(α+β)

If 1 < β < α then the order of the inequalities are reversed. For α, β > 1 the absolute distance between the mean and the median is less than 5% of the distance between the maximum and minimum values of x. On the other hand, the absolute distance between the mean and the mode can reach 50% of the distance between the maximum and minimum values of x, for the (pathological) case of α = 1 and β = 1, for which values the beta distribution approaches the uniform distribution and the differential entropy approaches its maximum value, and hence maximum "disorder".

For example, for α = 1.00001 and β = 1.000000001:

- mode = 0.9999; PDF(mode) = 1.00010
- mean = 0.500025; PDF(mean) = 1.00003
- median = 0.500035; PDF(median) = 1.00003
- mean - mode = -0.499875
- mean - median = -9.65538 × 10⁻⁹

where PDF stands for the value of the probability density function.



Mean, geometric mean and harmonic mean relationship

It is known from the inequality of arithmetic and geometric means that the geometric mean is lower than the mean. Similarly, the harmonic mean is lower than the geometric mean. The accompanying plot shows that for α = β, both the mean and the median are exactly equal to 1/2, regardless of the value of α = β, and the mode is also equal to 1/2 for α = β > 1, however the geometric and harmonic means are lower than 1/2 and they only approach this value asymptotically as α = β → ∞.

Kurtosis bounded by the square of the skewness

As remarked by Feller,[3] in the Pearson system the beta probability density appears as type I (any difference between the beta distribution and Pearson's type I distribution is only superficial and it makes no difference for the following discussion regarding the relationship between kurtosis and skewness). Karl Pearson showed, in Plate 1 of his paper [30] published in 1916, a graph with the kurtosis as the vertical axis (ordinate) and the square of the skewness as the horizontal axis (abscissa), in which a number of distributions were displayed.[29] The region occupied by the beta distribution is bounded by the following two lines in the (skewness^2,kurtosis) plane, or the (skewness^2,excess kurtosis) plane:

(skewness)^2 + 1 < kurtosis < (3/2)(skewness)^2 + 3
or, equivalently,
(skewness)^2 - 2 < excess kurtosis < 3/2(skewness)^2

At a time when there were no powerful digital computers, Karl Pearson accurately computed further boundaries,[31][32] for example, separating the "U-shaped" from the "J-shaped" distributions. The lower boundary line (excess kurtosis + 2 - skewness^2 = 0) is produced by skewed "U-shaped" beta distributions with both values of shape parameters α and β close to zero. The upper boundary line (excess kurtosis - (3/2) skewness^2 = 0) is also the intersection with Pearson's distribution III, which has unlimited support in one direction (towards positive infinity), and can be bell-shaped or J-shaped. His son, Egon Pearson, showed[33] that the region (in the kurtosis/squared-skewness plane) occupied by the beta distribution (equivalently, Pearson's distribution I) as it approaches this boundary (excess kurtosis - (3/2) skewness^2 = 0) is shared with the noncentral chi-squared distribution. Karl Pearson[33] (Pearson 1895, pp. 327, 366, 372-376) also showed that the gamma distribution is a Pearson type III distribution. Hence this boundary line (excess kurtosis + 2) is known as the gamma line. (This can be shown from the fact that the excess kurtosis of the gamma distribution is 6/k and the square of the skewness is 4/k, hence (excess kurtosis - (3/2) skewness^2 = 0) is identically satisfied by the gamma distribution regardless of the value of the parameter "k"). This is to be expected, since the chi-squared distribution, χ^2(k) is a special case of the gamma distribution, with parameterization X ~ Γ(k/2, 1/2) where k is a positive integer that specifies the "number of degrees of freedom" of the chi-squared distribution.

An example of a beta distribution near the upper boundary (excess kurtosis - (3/2) skewness^2 = 0) is given by α = 0.1, β = 1000, for which the ratio (excess kurtosis)/(skewness^2) = 1.49893 approaches the upper limit of 1.5 from below. An example of a beta distribution near the lower boundary (excess kurtosis + 2 - skewness^2 = 0) is given by α = 0.0001, β = 0.1, for which values the expression (excess kurtosis + 2)/(skewness^2) = 1.02621 approaches the lower limit of 1 from above. In the infinitesimal limit for both α and β approaching zero symmetrically, the excess kurtosis reaches its minimum value at -2. This minimum value occurs at the point at which the lower boundary line intersects the vertical axis (ordinate). (However, in Pearson's original chart, the ordinate is kurtosis, instead of excess kurtosis, and it increases downwards rather than upwards).

Values for the skewness and excess kurtosis below the lower boundary (excess kurtosis + 2 - skewness^2 = 0) cannot occur for any distribution, and hence Karl Pearson appropriately called the region below this boundary the "impossible region". The boundary for this "impossible region" is determined by (symmetric or skewed) bimodal U-shaped distributions for which the parameters α and β approach zero and hence all the probability density is concentrated at the ends: x = 0, 1 with practically nothing in between them. Since for α = β = 0 the probability density is concentrated at the two ends x = 0 and x = 1, this "impossible boundary" is determined by a Bernoulli distribution, where the two only possible outcomes occur with respective probabilities p and q = 1-p. For cases approaching this limit boundary with symmetry α = β, skewness = 0, excess kurtosis = -2 (this is the lowest excess kurtosis possible for any distribution), and the probabilities are p = q = 1/2. For cases approaching this limit boundary with skewness, excess kurtosis = -2 + skewness^2, and the probability density is concentrated more at one end than the other end (with practically nothing in between), with probabilities p = 1/(2+β) at the left end = 0 and q = 1-p = β/(2+β) at the right end = x = 1.

Symmetry

All statements are conditional on α, β > 0:

- Probability density function reflection symmetry
f(x; α, β) = f(1 - x; β, α)
- Cumulative distribution function reflection symmetry plus unitary translation
F(x; α, β) = I_α(α, β) = 1 - F(1 - x; β, α) = 1 - I_β-α(β, α)
- Mode reflection symmetry plus unitary translation
mode(B(α, β)) = 1 - mode(B(β, α)), if B(β, α) ≠ B(1, 1)
- Median reflection symmetry plus unitary translation
median(B(α, β)) = 1 - median(B(β, α))
- Mean reflection symmetry plus unitary translation
μ(B(α, β)) = 1 - μ(B(β, α))
- Geometric means each is individually asymmetric, the following symmetry applies between the geometric mean based on X and the geometric mean based on its reflection (1-X)
G_X(B(α, β)) = G_{1-X}(B(β, α))
- Harmonic means each is individually asymmetric, the following symmetry applies between the harmonic mean based on X and the harmonic mean based on its reflection (1-X)
H_X(B(α, β)) = H_{1-X}(B(β, α)) if α, β > 1.
- Variance symmetry
var(B(α, β)) = var(B(β, α))
- Geometric variances each is individually asymmetric, the following symmetry applies between the log geometric variance based on X and the log geometric variance based on its reflection (1-X)
ln(var_GX(B(α, β))) = ln(var_G(1-X)(B(β, α)))
- Geometric covariance symmetry
ln cov_G(α, β)(B(α, β)) = ln cov_G(1-X, 1-X)(B(β, α))
- Mean absolute deviation around the mean symmetry
E[|X - E[X]|](B(α, β)) = E[|X - E[X]|](B(β, α))
- Skewness skew-symmetry
skewness(B(α, β)) = -skewness(B(β, α))
- Excess kurtosis symmetry
excess kurtosis(B(α, β)) = excess kurtosis(B(β, α))
- Characteristic function symmetry of Real part (with respect to the origin of variable "t")
Re[E_1 F_1(α; α + β; it)] = Re[E_1 F_1(α; α + β - it)]
- Characteristic function skew-symmetry of Imaginary part (with respect to the origin of variable "t")
Im[E_1 F_1(α; α + β; it)] = -Im[E_1 F_1(α; α + β - it)]
- Characteristic function symmetry of Absolute value (with respect to the origin of variable "t")
Abs[E_1 F_1(α; α + β; it)] = Abs[E_1 F_1(α; α + β - it)]
- Differential entropy symmetry
h(B(α, β)) = h(B(β, α))
- Relative entropy (also called Kullback–Leibler divergence) symmetry
D_KL(X_1 || X_2) = D_KL(X_2 || X_1), if h(X_1) = h(X_2), for (skewed) α ≠ β
- Fisher information matrix symmetry
Z_β = Z_μ

Geometry of the probability density function

Inflection points
For certain values of the shape parameters α and β, the probability density function has inflection points, at which the curvature changes sign. The position of these inflection points can be useful as a measure of the dispersion or spread of the distribution.

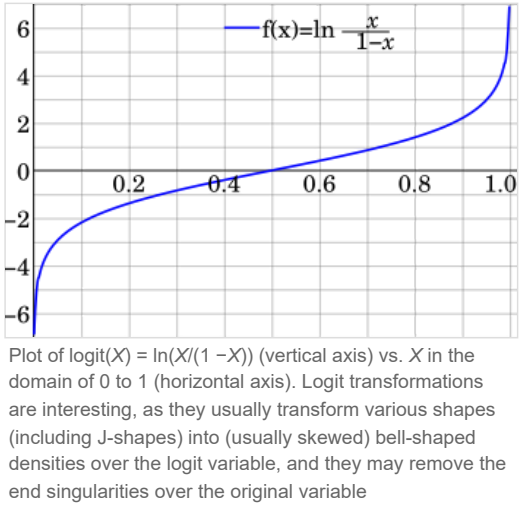
Defining the following quantity:

κ = (α-1)(β-1)/(α+β-2)

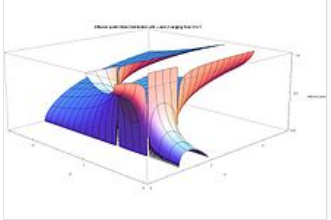
Points of inflection occur,[31][32][33] depending on the value of the shape parameters α and β, as follows:

- (α > 2, β > 2) The distribution is bell-shaped (symmetric for α = β and skewed otherwise), with two inflection points, equidistant from the mode:

z = mode ± κ = (α-1 ± √(α-1)(β-1))/(α+β-2)



Plot of logp(X) = ln(p(X)(1 - X)) (vertical axis) vs. X in the domain of 0 to 1 (horizontal axis). Logit transformations are interesting, as they usually transform various shapes (including J-shapes) into (usually skewed) bell-shaped densities over the logit variable, and they may remove the end singularities over the original variable.



Inflection point location versus α and β showing regions with one inflection point.

$$\begin{aligned}\ln{\mathcal{L}}(\alpha,\beta,a,c\mid Y) &= \sum_{i=1}^N \ln{\mathcal{L}}_i(\alpha,\beta,a,c\mid Y_i) \\ &= \sum_{i=1}^N \ln f(Y_i;\alpha,\beta,a,c) \\ &= \sum_{i=1}^N \ln \frac{(Y_i-a)^{\alpha-1}(c-Y_i)^{\beta-1}}{(c-a)^{\alpha+\beta-1}\mathrm{B}(\alpha,\beta)} \\ &= (\alpha-1)\sum_{i=1}^N \ln(Y_i-a) + (\beta-1)\sum_{i=1}^N \ln(c-Y_i) - N\ln\mathrm{B}(\alpha,\beta) - N(\alpha+\beta-1)\ln(c-a)\end{aligned}$$

Finding the maximum with respect to a shape parameter involves taking the partial derivative with respect to the shape parameter and setting the expression equal to zero yielding the **maximum likelihood estimator** of the shape parameters:

$$\begin{aligned}\frac{\partial \ln{\mathcal{L}}(\alpha,\beta,a,c\mid Y)}{\partial \alpha} &= \sum_{i=1}^N \ln(Y_i-a) - N(-\psi(\alpha+\beta) + \psi(\alpha)) - N\ln(c-a) = 0 \\ \frac{\partial \ln{\mathcal{L}}(\alpha,\beta,a,c\mid Y)}{\partial \beta} &= \sum_{i=1}^N \ln(c-Y_i) - N(-\psi(\alpha+\beta) + \psi(\beta)) - N\ln(c-a) = 0 \\ \frac{\partial \ln{\mathcal{L}}(\alpha,\beta,a,c\mid Y)}{\partial a} &= -(\alpha-1)\sum_{i=1}^N \frac{1}{Y_i-a} + N(\alpha+\beta-1)\frac{1}{c-a} = 0 \\ \frac{\partial \ln{\mathcal{L}}(\alpha,\beta,a,c\mid Y)}{\partial c} &= (\beta-1)\sum_{i=1}^N \frac{1}{c-Y_i} - N(\alpha+\beta-1)\frac{1}{c-a} = 0\end{aligned}$$

these equations can be re-arranged as the following system of four coupled equations (the first two equations are geometric means and the second two equations are the harmonic means) in terms of the maximum likelihood estimates for the four parameters **α
,
β
,
a
,
c
{\displaystyle \alpha ,\beta ,a,c}**:

$$\begin{aligned}\frac{1}{N}\sum_{i=1}^N \ln \frac{Y_i-a}{c-a} &= \psi(\hat{\alpha}) - \psi(\hat{\alpha}+\hat{\beta}) = \ln \hat{G}_X \\ \frac{1}{N}\sum_{i=1}^N \ln \frac{c-Y_i}{c-a} &= \psi(\hat{\beta}) - \psi(\hat{\alpha}+\hat{\beta}) = \ln \hat{G}_{1-X} \\ \frac{1}{\frac{1}{N}\sum_{i=1}^N \frac{c}{Y_i-a}} &= \frac{\hat{\alpha}-1}{\hat{\alpha}+\hat{\beta}-1} = \hat{H}_X \\ \frac{1}{\frac{1}{N}\sum_{i=1}^N \frac{c}{c-Y_i}} &= \frac{\hat{\beta}-1}{\hat{\alpha}+\hat{\beta}-1} = \hat{H}_{1-X}\end{aligned}$$

with sample geometric means:

$$\begin{aligned}\hat{G}_X &= \prod_{i=1}^N \left(\frac{Y_i-a}{c-a}\right)^{\frac{1}{N}} \\ \hat{G}_{1-X} &= \prod_{i=1}^N \left(\frac{c-Y_i}{c-a}\right)^{\frac{1}{N}}\end{aligned}$$

The parameters **α
,
β
{\displaystyle \alpha ,\beta }** are embedded inside the geometric mean expressions in a nonlinear way (to the power 1/N). This precludes, in general, a closed form solution, even for an initial value approximation for iteration purposes. One alternative is to use as initial values for iteration the values obtained from the method of moments solution for the four parameter case. Furthermore, the expressions for the harmonic means are well-defined only for **α
,
β
>
1
{\displaystyle \alpha ,\beta >1}**, which precludes a maximum likelihood solution for shape parameters less than unity in the four-parameter case. Fisher's information matrix for the four parameter case is positive-definite only for

α
,
β
>
2
{\displaystyle \alpha ,\beta >2}

 (for further discussion, see section on Fisher information matrix, four parameter case), for bell-shaped (symmetric or unsymmetric) beta distributions, with inflection points located to either side of the mode. The following Fisher information components (that represent the expectations of the curvature of the log likelihood function) have *singularities* at the following values:

$$\begin{array}{ll}\alpha = 2: & \mathbb{E}\left[\frac{1}{N}\frac{\partial^2 \ln{\mathcal{L}}(\alpha,\alpha,c\mid Y)}{\partial \alpha^2}\right] = \mathcal{I}_{\alpha\alpha} \\ \beta = 2: & \mathbb{E}\left[\frac{1}{N}\frac{\partial^2 \ln{\mathcal{L}}(\alpha,\beta,c\mid Y)}{\partial \beta^2}\right] = \mathcal{I}_{\beta\beta} \\ \alpha = 2: & \mathbb{E}\left[\frac{1}{N}\frac{\partial^2 \ln{\mathcal{L}}(\alpha,\beta,a,c\mid Y)}{\partial \alpha \partial \alpha}\right] = \mathcal{I}_{\alpha\alpha} \\ \beta = 1: & \mathbb{E}\left[\frac{1}{N}\frac{\partial^2 \ln{\mathcal{L}}(\alpha,\beta,a,c\mid Y)}{\partial \beta \partial \beta}\right] = \mathcal{I}_{\beta\beta}\end{array}$$

(for further discussion see section on Fisher information matrix). Thus, it is not possible to strictly carry on the maximum likelihood estimation for some well known distributions belonging to the four-parameter beta distribution family, like the uniform distribution (Beta(1, 1, a, c)), and the arcsine distribution (Beta(1/2, 1/2, a, c)). N.L.Johnson and S.Kotz^[1] ignore the equations for the harmonic means and instead suggest "If a and c are unknown, and maximum likelihood estimators of α, c, α and β are required, the above procedure (for the two unknown parameter case, with X transformed as Y = (Y − a)/(c − a)) can be repeated using a succession of trial values of a and c, until the pair (a, c) for which maximum likelihood (given a and c) is as great as possible, is attained" (where, for the purpose of clarity, their notation for the parameters has been translated into the present notation).

Fisher information matrix

Let a random variable *X* have a probability density *f*(*X*;*α*). The partial derivative with respect to the (unknown, and to be estimated) parameter *α* of the log likelihood function is called the **score**. The second moment of the score is called the Fisher information:

$$\mathcal{I}(\alpha)=\mathbb{E}\left[\left(\frac{\partial}{\partial \alpha}\ln{\mathcal{L}}(\alpha\mid X)\right)^2\right].$$

The expectation of the score is zero, therefore the Fisher information is also the second moment centered on the mean of the score: the variance of the score.

If the **log likelihood function** is twice differentiable with respect to the parameter *α*, and under certain regularity conditions,^[a] then the Fisher information may also be written as follows (which is often a more convenient form for calculation purposes):

$$\mathcal{I}(\alpha)=-\mathbb{E}\left[\frac{\partial^2}{\partial \alpha^2}\ln{\mathcal{L}}(\alpha\mid X)\right].$$

Thus, the Fisher information is the negative of the expectation of the second derivative with respect to the parameter *α* of the log likelihood function. Therefore, Fisher information is a measure of the curvature of the log likelihood function of *α*. A low curvature (and therefore high radius of curvature), flatter log likelihood function curve has low Fisher information; while a log likelihood function curve with large curvature (and therefore low radius of curvature) has high Fisher information. When the Fisher information matrix is computed at the evaluates of the parameters ("the observed Fisher information matrix") it is equivalent to the replacement of the true log likelihood surface by a Taylor's series approximation, taken as far as the quadratic terms.^[a] The word information, in the context of Fisher information, refers to information about the parameters. Information such as estimation, sufficiency and properties of variances of estimators. The Cramér–Rao bound states that the inverse of the Fisher information is a lower bound on the variance of any estimator of a parameter *α*:

$$\mathrm{var}[\hat{\alpha}]\geq \frac{1}{\mathcal{I}(\alpha)}.$$

The precision to which one can estimate the estimator of a parameter *α* is limited by the Fisher Information of the log likelihood function. The Fisher information is a measure of the minimum error involved in estimating a parameter of a distribution and it can be viewed as a measure of the resolving power of an experiment needed to discriminate between two alternative hypothesis of a parameter.^[a]

When there are *N* parameters

$$\begin{bmatrix}\theta_1\\\theta_2\\\vdots \\\theta_N\end{bmatrix},$$

then the Fisher information takes the form of an *N* × *N* positive semidefinite symmetric matrix, the Fisher information matrix, with typical element:

$$\mathcal{I}(\theta)_\mu\nu=\mathbb{E}\left[\left(\frac{\partial}{\partial \theta_\mu}\ln{\mathcal{L}}\right)\left(\frac{\partial}{\partial \theta_\nu}\ln{\mathcal{L}}\right)\right].$$

Under certain regularity conditions,^[a] the Fisher Information Matrix may also be written in the following form, which is often more convenient for computation:

$$\mathcal{I}(\theta)_\mu\nu=-\mathbb{E}\left[\frac{\partial^2}{\partial \theta_\mu\partial \theta_\nu}\ln{\mathcal{L}}\right].$$

With *X*₁, ..., *X*_{*N*} iid random variables, an *N*-dimensional "box" can be constructed with sides *X*₁, ..., *X*_{*N*}. Costa and Cover^[a] show that the (Shannon) differential entropy *h*(*X*) is related to the volume of the typical set (having the sample entropy close to the true entropy), while the Fisher information is related to the surface of this typical set.

Two parameters

For *X*₁, ..., *X*_{*N*} independent random variables each having a beta distribution parametrized with shape parameters *α* and *β*, the joint log likelihood function for *N* iid observations is:

$$\ln{\mathcal{L}}(\alpha,\beta\mid X)= (\alpha-1)\sum_{i=1}^N \ln X_i + (\beta-1)\sum_{i=1}^N \ln(1-X_i) - N\ln\mathrm{B}(\alpha,\beta)$$

therefore the joint log likelihood function per *N* iid observations is

$$\frac{1}{N}\ln{\mathcal{L}}(\alpha,\beta\mid X)= (\alpha-1)\frac{1}{N}\sum_{i=1}^N \ln X_i + (\beta-1)\frac{1}{N}\sum_{i=1}^N \ln(1-X_i) - \ln\mathrm{B}(\alpha,\beta).$$

For the two parameter case, the Fisher information has 4 components: 2 diagonal and 2 off-diagonal. Since the Fisher information matrix is symmetric, one of these off diagonal components is independent. Therefore, the Fisher information matrix has 3 independent components (2 diagonal and 1 off diagonal).

Aryal and Nadarajah^[5] calculated Fisher's information matrix for the four-parameter case, from which the two parameter case can be obtained as follows:

$$\begin{aligned}\frac{\partial^2 \ln{\mathcal{L}}(\alpha,\beta\mid X)}{\partial \alpha^2} &= \psi_1[\ln(X)] = \psi_1(\alpha) - \psi_1(\alpha+\beta) = \mathcal{I}_{\alpha\alpha} = \mathbb{E}\left[\frac{\partial^2 \ln{\mathcal{L}}(\alpha,\beta\mid X)}{N\partial \alpha^2}\right] = \ln\mathrm{var}_{\mathcal{L}}X \\ \frac{\partial^2 \ln{\mathcal{L}}(\alpha,\beta\mid X)}{N\partial \beta^2} &= \mathrm{var}[\ln(1-X)] = \psi_1(\beta) - \psi_1(\alpha+\beta) = \mathcal{I}_{\beta\beta} = \mathbb{E}\left[\frac{\partial^2 \ln{\mathcal{L}}(\alpha,\beta\mid X)}{N\partial \beta^2}\right] = \ln\mathrm{var}_{\mathcal{L}}(1-X) \\ \frac{\partial^2 \ln{\mathcal{L}}(\alpha,\beta\mid X)}{N\partial \alpha \partial \beta} &= \mathrm{cov}[\ln X, \ln(1-X)] = -\psi_1(\alpha+\beta) = \mathcal{I}_{\alpha\beta} = \mathbb{E}\left[\frac{\partial^2 \ln{\mathcal{L}}(\alpha,\beta\mid X)}{N\partial \alpha \partial \beta}\right] = \ln\mathrm{cov}_{\mathcal{L}}(X,1-X)\end{aligned}$$

Since the Fisher information matrix is symmetric

$$\mathcal{I}_{\alpha\beta} = \mathcal{I}_{\beta\alpha} = \ln\mathrm{cov}_{\mathcal{L}}(X,1-X)$$

The Fisher information components are equal to the log geometric variances and log geometric covariance. Therefore, they can be expressed as **trigamma functions**, denoted

ψ
1
(
α
)

{\displaystyle \psi _{1}(\alpha)}

, the second of the **polygamma functions**, defined as the derivative of the **digamma** function:

$$\psi _1(\alpha)={\frac {\mathrm {d} ^{2}\ln \Gamma (\alpha)}{\mathrm {d} \alpha ^{2}}}={\frac {\partial ^{2}\psi (\alpha)}{\partial \alpha ^{2}}}.$$

These derivatives are also derived in the § Two unknown parameters and plots of the log likelihood function are also shown in that section. § Geometric variance and covariance contains plots and further discussion of the Fisher information matrix components: the log geometric variances and log geometric covariance as a function of the shape parameters *α* and *β*. § Moments of logarithmically transformed random variables contains formulas for moments of logarithmically transformed random variables. Images for the Fisher information components **I

α
α

,

I

β
β

and

I

α
β

{\displaystyle I_{\alpha \alpha },I_{\beta \beta }andI_{\alpha \beta }}** are shown in § Geometric variance.

The determinant of Fisher's information matrix is of interest (for example for the calculation of *Jeffreys prior* probability). From the expressions for the individual components of the Fisher information matrix, it follows that the determinant of Fisher's (symmetric) information matrix for the beta distribution is:

$$\begin{aligned}\det(\mathcal{I}(\alpha,\beta)) &= \mathcal{I}_{\alpha\alpha}\mathcal{I}_{\beta\beta} - \mathcal{I}_{\alpha\beta}\mathcal{I}_{\beta\alpha} \\ &= (\psi_1(\alpha) - \psi_1(\alpha+\beta))(\psi_1(\beta) - \psi_1(\alpha+\beta)) - (-\psi_1(\alpha+\beta))(-\psi_1(\alpha+\beta)) \\ &= \psi_1(\alpha)\psi_1(\beta) - (\psi_1(\alpha) + \psi_1(\beta))\psi_1(\alpha+\beta)\end{aligned}$$

$$\lim_{\alpha\rightarrow 0}\det(\mathcal{I}(\alpha,\beta)) = \lim_{\alpha\rightarrow 0}\det(\mathcal{Z}(\alpha,\beta)) = 0$$

$$\lim_{\beta\rightarrow 0}\det(\mathcal{I}(\alpha,\beta)) = \lim_{\beta\rightarrow 0}\det(\mathcal{Z}(\alpha,\beta)) = 0$$

From *Sylvester's criterion* (checking whether the diagonal elements are all positive), it follows that the Fisher information matrix for the two parameter case is **positive-definite** (under the standard condition that the shape parameters are positive *α* > 0 and *β* > 0).

Four parameters

If *Y*₁, ..., *Y*_{*N*} are independent random variables each having a beta distribution with four parameters: the exponents *α* and *β*, and also *a* (the minimum of the distribution range), and *c* (the maximum of the distribution range) (section titled "Alternative parametrizations", "Four parameters"), with **probability density function**:

$$f(Y;\alpha,\beta,a,c) = \frac{f(\alpha,\alpha,\beta)}{c-a} = \frac{\left(\frac{Y-a}{c-a}\right)^{\alpha-1}\left(\frac{c-Y}{c-a}\right)^{\beta-1}}{(c-a)\mathrm{B}(\alpha,\beta)} = \frac{(y-a)^{\alpha-1}(c-y)^{\beta-1}}{(c-a)^{\alpha+\beta-1}\mathrm{B}(\alpha,\beta)}.$$

the joint log likelihood function per *N* iid observations is:

$$\frac{1}{N}\ln{\mathcal{L}}(\alpha,\beta,a,c\mid Y) = \frac{\alpha-1}{N}\sum_{i=1}^N \ln(Y_i-a) + \frac{\beta-1}{N}\sum_{i=1}^N \ln(c-Y_i) - \ln\mathrm{B}(\alpha,\beta) - (\alpha+\beta-1)\ln(c-a)$$

For the four parameter case, the Fisher information has 4*4=16 components. It has 12 off-diagonal components = (4×4 total − 4 diagonal). Since the Fisher information matrix is symmetric, half of these components (12÷2=6) are independent. Therefore, the Fisher information matrix has 6 independent off-diagonal + 4 diagonal = 10 independent components. Aryal and Nadarajah^[5] calculated Fisher's information matrix for the four parameter case as follows:

$$\begin{aligned}\frac{1}{N}\frac{\partial^2 \ln{\mathcal{L}}(\alpha,\beta,a,c\mid Y)}{\partial \alpha^2} &= \mathrm{var}[\ln(X)] = \psi_1(\alpha) - \psi_1(\alpha+\beta) = \mathcal{I}_{\alpha\alpha} = \mathbb{E}\left[\frac{1}{N}\frac{\partial^2 \ln{\mathcal{L}}(\alpha,\beta,a,c\mid Y)}{\partial \alpha^2}\right] = \ln(\mathrm{var}_{\mathcal{L}}X) \\ \frac{1}{N}\frac{\partial^2 \ln{\mathcal{L}}(\alpha,\beta,a,c\mid Y)}{\partial \beta^2} &= \mathrm{var}[\ln(1-X)] = \psi_1(\beta) - \psi_1(\alpha+\beta) = \mathcal{I}_{\beta\beta} = \mathbb{E}\left[\frac{1}{N}\frac{\partial^2 \ln{\mathcal{L}}(\alpha,\beta,a,c\mid Y)}{\partial \beta^2}\right] = \ln(\mathrm{var}_{\mathcal{L}}(1-X)) \\ \frac{1}{N}\frac{\partial^2 \ln{\mathcal{L}}(\alpha,\beta,a,c\mid Y)}{\partial \alpha \partial \beta} &= \mathrm{cov}[\ln X, \ln(1-X)] = -\psi_1(\alpha+\beta) = \mathcal{I}_{\alpha\beta} = \mathbb{E}\left[\frac{1}{N}\frac{\partial^2 \ln{\mathcal{L}}(\alpha,\beta,a,c\mid Y)}{\partial \alpha \partial \beta}\right] = \ln(\mathrm{cov}_{\mathcal{L}}(X,1-X))\end{aligned}$$

In the above expressions, the use of *X* instead of *Y* in the expressions

var
⁡
[
ln
⁡
(
X
)
]
=
ln
⁡
(
var
⁡
X
)

{\displaystyle \mathrm {var} [\ln(X)]=\ln(\mathrm {var} X)}

 is *not an error*. The expressions in terms of the log geometric variances and log geometric covariance occur as functions of the two parameter *X* = Beta(*α*,*β*) parametrization because when taking the partial derivatives with respect to the exponents *α*, *β* in the four parameter case, one obtains the identical expressions as for the two parameter case; these terms of the four parameter Fisher information matrix are independent of the minimum *a* and maximum *c* of the distribution's range. The only non-zero variable *ln*(*c* − *y*) (with domain, *a* < *y*) is equivalent to the *Haldane prior* on the domain [0, 1] was pointed out by Harold Jeffreys in the first edition (1939) of his book *Theory of Probability* (^[a] p. 143). Jeffreys writes "Certainly if we take the *Bayes–Laplace* rule right up to the extreme we are led to results that do not correspond to anybody's way of thinking. The (*Haldane*) rule

d
y
/
(
c
−
y
)

{\displaystyle dy/(c-y)}

 goes too far in the other way. It would lead to the conclusion that if a sample is of one type with respect to some property there is a probability ... that the whole population is of that type." The fact that that we are always at the end of an average run, and more reasonably to assume that we are in the middle of an average run. Clearly a higher value for both probabilities is necessary if they are to accord with reasonable belief". These problems with Laplace's rule of succession motivated Haldane, Perks, Jeffreys and others to search for other forms of prior probability (see the next § Bayesian inference). According to Jaynes,^[5] the main problem with the rule of succession is that it is not valid when *s*=0 or *s*=*n* (see rule of succession, for an analysis of its validity).

The Fisher information for *N* iid samples is *N* times the individual Fisher information (eq. 1.279, page 394 of Cover and Thomas^[a]). (Aryal and Nadarajah^[5] take a single observation, *N* = 1, to calculate the following components of the Fisher information, which leads to the same result as considering the derivatives of the log likelihood per *N* observations. Moreover, below the erroneous expression for **I

α
α

{\displaystyle I_{\alpha \alpha }}** in Aryal and Nadarajah has been corrected.)

$$\begin{array}{ll}\alpha > 2: & \mathbb{E}\left[\frac{1}{N}\frac{\partial^2 \ln{\mathcal{L}}(\alpha,\beta,a,c\mid Y)}{\partial \alpha^2}\right] = \mathcal{I}_{\alpha\alpha} = \frac{\beta(\alpha+\beta-1)}{(\alpha-2)(c-a)^2} \\ \beta > 2: & \mathbb{E}\left[\frac{1}{N}\frac{\partial^2 \ln{\mathcal{L}}(\alpha,\beta,a,c\mid Y)}{\partial \beta^2}\right] = \mathcal{I}_{\beta\beta} = \frac{\alpha(\alpha+\beta-1)}{(\beta-2)(c-a)^2} \\ & \mathbb{E}\left[\frac{1}{N}\frac{\partial^2 \ln{\mathcal{L}}(\alpha,\beta,a,c\mid Y)}{\partial \alpha \partial c}\right] = \mathcal{I}_{\alpha c} = \frac{(\alpha+\beta-1)}{(c-a)^2} \\ \alpha > 1: & \mathbb{E}\left[\frac{1}{N}\frac{\partial^2 \ln{\mathcal{L}}(\alpha,\beta,a,c\mid Y)}{\partial \alpha \partial \alpha}\right] = \mathcal{I}_{\alpha\alpha} = \frac{\beta}{(\alpha-1)(c-a)} \\ & \mathbb{E}\left[\frac{1}{N}\frac{\partial^2 \ln{\mathcal{L}}(\alpha,\beta,a,c\mid Y)}{\partial \alpha \partial c}\right] = \mathcal{I}_{\alpha c} = \frac{1}{(c-a)} \\ & \mathbb{E}\left[\frac{1}{N}\frac{\partial^2 \ln{\mathcal{L}}(\alpha,\beta,a,c\mid Y)}{\partial \beta \partial \alpha}\right] = \mathcal{I}_{\beta\alpha} = -\frac{1}{(c-a)} \\ \beta > 1: & \mathbb{E}\left[\frac{1}{N}\frac{\partial^2 \ln{\mathcal{L}}(\alpha,\beta,a,c\mid Y)}{\partial \beta \partial \beta}\right] = \mathcal{I}_{\beta\beta} = -\frac{\alpha}{(\beta-1)(c-a)}\end{array}$$

The lower two diagonal entries of the Fisher information matrix, with respect to the parameter *a* (the minimum of the distribution's range): **I

α
α

{\displaystyle I_{\alpha \alpha }}** and with respect to the parameter *c* (the maximum of the distribution's range): **I

β
β

{\displaystyle I_{\beta \beta }}** are only defined for exponents *α* > 2 and *β* > 2 respectively. The Fisher information matrix component **I

α
a

{\displaystyle I_{\alpha a}}** for the minimum *a* approaches infinity for exponent *a* approaching *z* from above, and the Fisher information matrix component **I

a
c

{\displaystyle I_{ac}}** for the maximum *c* approaches infinity for exponent *β* approaching *z* from above.

The Fisher information matrix for the four parameter case does not depend on the individual values of the minimum *a* and the maximum *c*, but only on the total range (*c* − *a*). Moreover, the components of the Fisher information matrix that depend on the range (*c* − *a*), depend only through its inverse (or the square of the inverse), such that the Fisher information decreases for increasing range (*c* − *a*).

The accompanying images show the Fisher information components **I

α
α

and

I

α
a

{\displaystyle I_{\alpha \alpha }andI_{\alpha a}}** images for the Fisher information components **I

α
a

and

I

a
β

{\displaystyle I_{\alpha a}andI_{a\beta }}** are shown in § Geometric variance. The three Fisher information components look like a basin, with the "walls" of the basin being located at low values of the parameters.

The following four-parameter-beta-distribution Fisher information components can be expressed in terms of the two-parameter, *X* = Beta(*α*,*β*) expectations of the transformed ratio (1 − *X*)/*X* and of its mirror image

X
/
(
1
−
X
)

{\displaystyle X/(1-X)}

 and of its mirror image

X
/
(
1
−
X
)

{\displaystyle X/(1-X)}

, scaled by the range (*c* − *a*), which may be helpful for interpretation:

$$\begin{aligned}\mathcal{I}_{\alpha a} &= -\frac{\mathbb{E}\left[\frac{1-X}{X}\right]}{c-a} = -\frac{\beta}{(\alpha-1)(c-a)}\text{ if }\alpha > 1 \\ \mathcal{I}_{\beta c} &= -\frac{\mathbb{E}\left[\frac{X}{1-X}\right]}{c-a} = -\frac{\alpha}{(\beta-1)(c-a)}\text{ if }\beta > 1\end{aligned}$$

These are also the expected values of the "inverted beta distribution" or *beta prime distribution* (also known as beta distribution of the second kind or Pearson's Type IV)^[5] and its mirror image, scaled by the range (*c* − *a*).

Also, the following Fisher information components can be expressed in terms of the harmonic (*s*/*N*) variances or of variances based on the ratio transformed variables ((1-*X*)/*X*) as follows:

$$\begin{aligned}\alpha > 2: & \mathcal{I}_{\alpha a} = \mathrm{var}\left[\frac{1}{X}\left(\frac{\alpha-1}{c-a}\right)^2\right] = \mathrm{var}\left[\frac{1-X}{X}\left(\frac{\alpha-1}{c-a}\right)^2\right] = \frac{\beta(\alpha+\beta-1)}{(\alpha-2)(c-a)^2} \\ \beta > 2: & \mathcal{I}_{\beta c} = \mathrm{var}\left[\frac{1}{1-X}\left(\frac{\beta-1}{c-a}\right)^2\right] = \mathrm{var}\left[\frac{1}{1-X}\left(\frac{\beta-1}{c-a}\right)^2\right] = \frac{\alpha(\alpha+\beta-1)}{(\beta-2)(c-a)^2} \\ & \mathcal{I}_{\alpha c} = \mathrm{cov}\left[\frac{1}{X},\frac{1}{1-X}\right](\alpha-1)(\beta-1) = \mathrm{cov}\left[\frac{1-X}{X},\frac{X}{1-X}\right](\alpha-1)(\beta-1) = \frac{\alpha+\beta-1}{(c-a)^2}\end{aligned}$$

See section "Moments of linearly transformed, product and inverted random variables" for these expectations.

The determinant of Fisher's information matrix is of interest (for example for the calculation of *Jeffreys prior* probability). From the expressions for the individual components, it follows that the determinant of Fisher's (symmetric) information matrix for the beta distribution with four parameters is:

$$\begin{aligned}\det(\mathcal{I}(\alpha,\beta,a,c)) &= -\mathcal{I}_{\alpha a}^2\mathcal{I}_{\alpha\beta} + \mathcal{I}_{\alpha a}\mathcal{I}_{\alpha c}\mathcal{I}_{\beta\beta} + \mathcal{I}_{\alpha\beta}^2\mathcal{I}_{\beta c} - \mathcal{I}_{\alpha a}\mathcal{I}_{\beta c}\mathcal{I}_{\alpha\beta}^2 \\ &\quad - \mathcal{I}_{\alpha a}\mathcal{I}_{\alpha c}\mathcal{I}_{\beta\beta} + \mathcal{I}_{\alpha\beta}^2\mathcal{I}_{\alpha c}\mathcal{I}_{\beta\beta} + \mathcal{I}_{\alpha c}^2\mathcal{I}_{\alpha\beta}\mathcal{I}_{\beta\beta} \\ &\quad - 2\mathcal{I}_{\alpha a}\mathcal{I}_{\alpha c}\mathcal{I}_{\beta\beta} + \mathcal{I}_{\alpha\beta}^2\mathcal{I}_{\beta c}^2 - \mathcal{I}_{\alpha a}\mathcal{I}_{\beta c}\mathcal{I}_{\alpha\beta}^2 + \mathcal{I}_{\alpha c}\mathcal{I}_{\beta c}^2\mathcal{I}_{\beta\beta} \\ &\quad - \mathcal{I}_{\alpha a}\mathcal{I}_{\alpha c}\mathcal{I}_{\beta\beta} - \mathcal{I}_{\alpha a}\mathcal{I}_{\alpha\beta}\mathcal{I}_{\beta c} + \mathcal{I}_{\alpha\beta}^2\mathcal{I}_{\alpha c}\mathcal{I}_{\beta c} \\ &\quad - \mathcal{I}_{\alpha a}\mathcal{I}_{\alpha\beta}\mathcal{I}_{\beta\beta} + \mathcal{I}_{\alpha c}\mathcal{I}_{\alpha\beta}\mathcal{I}_{\beta c} - \mathcal{I}_{\alpha c}\mathcal{I}_{\beta c}^2\mathcal{I}_{\beta\beta} \\ &\quad + 2\mathcal{I}_{\alpha a}\mathcal{I}_{\alpha\beta}\mathcal{I}_{\beta c} - \mathcal{I}_{\alpha a}\mathcal{I}_{\beta c}^2\mathcal{I}_{\beta\beta} + \mathcal{I}_{\alpha c}\mathcal{I}_{\alpha\beta}\mathcal{I}_{\beta c}\text{ if }\alpha,\beta > 2\end{aligned}$$

Using Sylvester's criterion (checking whether the diagonal elements are all positive), and since diagonal components **I

α
a

{\displaystyle I_{\alpha a}}** and **I

a
c

{\displaystyle I_{ac}}** have singularities at *α*=2 and *β*=2 it follows that the Fisher information matrix for the four parameter case is positive-definite for *α*>2 and *β*>2. Since for *α* > 2 and *β* > 2 the beta distribution is (symmetric or unsymmetric) bell shaped, it follows that the Fisher information matrix is positive-definite only for bell-shaped (symmetric or unsymmetric) beta distributions, with inflection points located to either side of the mode. Thus, important well known distributions belonging to the four-parameter beta distribution family, like the parabolic distribution (Beta(2,2,*a*,*c*)) and the uniform distribution (Beta(1,1,*a*,*c*)) have Fisher information components (**I

α
a

,

I

a
c

,

I

α
a
β

{\displaystyle I_{\alpha a},I_{ac},I_{\alpha \beta }}**) that blow up (approach infinity) in the four-parameter case (although their Fisher information components are all defined for the two parameter case). The four-parameter Wigner semicircle distribution (Beta(3/2,3/2,*a*,*c*)) and arcsine distribution (Beta(1/2,1/2,*a*,*c*)) have negative Fisher information determinants for the four-parameter case.

Bayesian inference

The use of Beta distributions in Bayesian inference is due to the fact that they provide a family of *conjugate prior* probability distributions for binomial (including Bernoulli) and geometric distributions. The domain of the beta distribution can be viewed as a probability, and in fact the beta distribution is often used to describe the distribution of a probability value *p*.^[a]

$$P(p;\alpha,\beta) = \frac{p^{\alpha-1}(1-p)^{\beta-1}}{\mathrm{B}(\alpha,\beta)}.$$

Examples of beta distributions used as prior probabilities to represent ignorance of prior parameter values in Bayesian inference are Beta(1,1), Beta(*α*,0) and Beta(1/2,1/2).

Rule of succession</

